BC 3 Name: _____
Vectors 2

1. If $\vec{v} = 2\vec{i} - 6\vec{j}$, find each of the follow.

   a. $-3\vec{v}$ b. $\left|\vec{v}\right|$ c. $\left|4\vec{v}\right|$

   d. a unit vector in the direction of $\vec{v}$.

2. Given points $A(-3, 5)$ and $B(1, 3)$, find $\overrightarrow{AB}$.

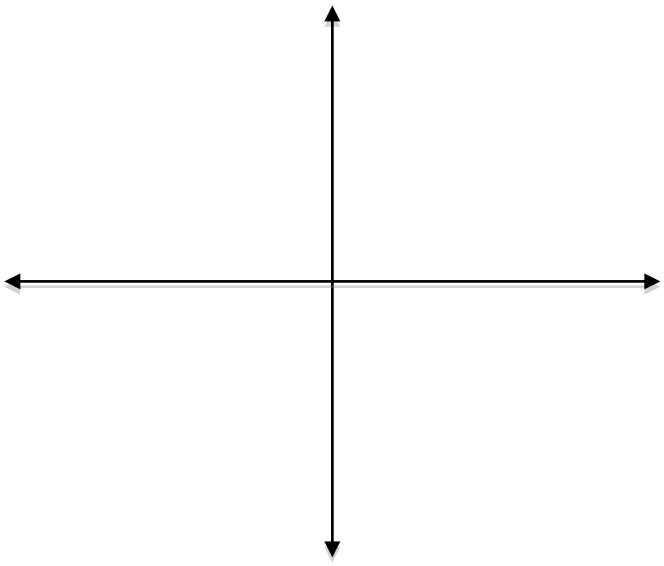
3. Given $\vec{v} = \langle -1, 4 \rangle$, find the unit vector in the direction of $\vec{v}$ and the angle that $\vec{v}$ makes with the positive x-axis.

In the plane, consider a position vector given by $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$. Usually, we draw these vectors for various values of t with their tails at the origin, so that their heads are at $\left(x(t), y(t)\right)$, the position of the object at time t. The velocity and acceleration vectors, respectively, are given by

$\vec{v}(t) = \vec{r}\,'(t) = \dfrac{dx}{dt}\vec{i} + \dfrac{dy}{dt}\vec{j}$ and $\vec{a}(t) = \vec{v}\,'(t) = \vec{r}\,''(t) = \dfrac{d^2x}{dt^2}\vec{i} + \dfrac{d^2y}{dt^2}\vec{j}$.

4. Given the position function $\vec{r}(t) = \left(e^t\right)\vec{i} + \left(\dfrac{2}{9}e^{2t}\right)\vec{j}$, find its velocity and acceleration vectors when $t = \ln(3)$. Please simplify your final results completely.

5. A particle moves on the plane so that its position at time t is given by the position vector $\vec{r}(t) = \langle 3\cos(t),\ 2\sin(t) \rangle$. Sketch the graph of the path traced out by the particle. Find both $\vec{v}\left(\frac{\pi}{4}\right)$ and $\vec{a}\left(\frac{\pi}{4}\right)$. Sketch the vectors $\vec{r}\left(\frac{\pi}{4}\right)$, $\vec{v}\left(\frac{\pi}{4}\right)$, and $\vec{a}\left(\frac{\pi}{4}\right)$. Find the speed of the particle at $t = \frac{\pi}{4}$. Usually, we draw the vectors $\vec{v}$ and $\vec{a}$ with their tails at the position of the object (particle) on the graph. (Note: The velocity vector should be tangent to the path of the object in the direction of the orientation. This is good!)



6. If $f(x) = x^3 - 2x + 3$, find a unit tangent vector and a unit normal vector to the graph of $f(x)$ at the point where $x = -2$.

7. Determine each of the following integrals

a. $\int \langle \cos(3t),\ -\sin(3t) \rangle \, dt$

b. $\int_0^1 \left(t^2 \vec{i} + t^3 \vec{j} \right) dt$

8. Find the position function $\vec{r}(t)$ for each acceleration vector and initial conditions.

a. $\vec{a}(t) = -3t\vec{i}$, $\vec{v}(0) = 2\vec{j}$ and $\vec{r}(0) = 4\vec{i}$

b. $\vec{a}(t) = \frac{1}{(1+t)}\vec{i} - e^{-t}\vec{j}$, $\vec{v}(0) = -\vec{i} + \vec{j}$ and $\vec{r}(0) = \frac{1}{3}\vec{i} - \vec{j}$

c. $\vec{a}(t) = \langle t + \cos(2t),\ 3t \rangle$, $\vec{v}(0) = \langle 1,\ 2 \rangle$ and $\vec{r}(0) = \langle 0,\ 0 \rangle$